

Bonus Quiz Week 7

April 12, 2013

1. Let $G = \{0 | \downarrow + \star\}$. Is $G > 0$, $G < 0$, $G = 0$, or $G || 0$?

Proof. If Left goes first, Left moves to 0 and wins. If Right goes first, Right moves to $\downarrow + \star$ which we know from lecture is fuzzy with 0. Thus, since it is Left's move, Left wins. Since Left wins no matter who goes first, $G > 0$. □

2. Find the Grundy number $G(5)$ for the subtraction game $\{1, 2, 4\}$

Proof. We know $G(0) = 0$, $G(1) = \text{mex}\{G(0)\} = \text{mex}0 = 1$, and continuing this we get

0	1	2	3	4	5
0	1	2	0	1	2

so $G(5) = 2$. □

3. $5 + \star$ is:

Proof. $\star = \{0 | 0\}$ and $5 = \{4 | \}$ so the Left options are to move to $0 + 5$ or $4 + \star$ and the right option is to move to $0 + 5$. Thus, $5 + \star = \{4 + \star, 5 | 5\}$ but $4 + \star$ is dominated by 5 so $5 + \star = \{4 + \star, 5 | 5\} = \{5 | 5\}$. □

4. What number is $\{\frac{1}{2} + \uparrow | 1\}$?

Proof. We know that $\frac{1}{2} + \uparrow > \frac{1}{2}$, but we also know that $\uparrow < \frac{1}{2^n}$ so $\frac{1}{2} + \uparrow < 1$. In particular, $\uparrow < \frac{1}{4}$ so $\frac{1}{2} + \uparrow < \frac{3}{4} < 1$ so $\{\frac{1}{2} + \uparrow | 1\} = \frac{3}{4}$. □

5. What size nim heap is equivalent to the sum $\star 2 + \star 6 + \star 5$?

Proof. First write out each number in binary and take the sum without carry:

$$\begin{array}{rcl} 2 & = & 010 \\ 6 & = & 110 \\ 5 & = & \underline{101} \\ & & 001 \end{array}$$

Converting back from binary we have $001 = 1$ so the nimsum of 2, 6, and 5 is 1. □